Nguyễn Trọng Đạt

Computer Engineering Student
Robotics ▪ AI ▪ Drones ▪ Embedded Systems

Ho Chi Minh City, Vietnam ▪ (+84) 399 922 395
trongdat4work@gmail.com ▪ github.com/LemonKronos ▪ linkedin.com/in/trong-dat



Education

Ho Chi Minh University of Technology (HCMUT – BKU)
B.Eng. in Computer Engineering
Expected graduation 2027
GPA: 2.9/4.0

Coursework: Embedded Systems, Operating Systems, Machine Learning, Microprocessors & Microcontrollers, IoT-Oriented Logic Design, AI-Oriented Multidisciplinary Project

Skills

Programming
C++, Python, Rust
also C, Shell script;
familiar with Java, Go, Zig

Robotics
ROS2 (Jazzy), Gazebo Harmonic, PX4 Autopilot, Micro XRCE-DDS

AI / ML
Reinforcement Learning, Imitation Learning, Latent Diffusion Learning, PyTorch, robotic data processing, AI edge computing

Tools
Git, Linux, VS Code

Languages

English – TOEIC L&R 970/990
Vietnamese – native

Interests

Drones, open source, and the Linux community. Long-term goal: Robotics Engineer.

Summary

Computer Engineering student focused on robotics, embedded systems, and autonomous drone software. Experienced with **ROS2, PX4, Gazebo, and C++** for real-time systems and simulation, and comfortable contributing safely to shared Git codebases. Seeking a part-time engineering role, with the option of moving to full-time after graduation.

Experience

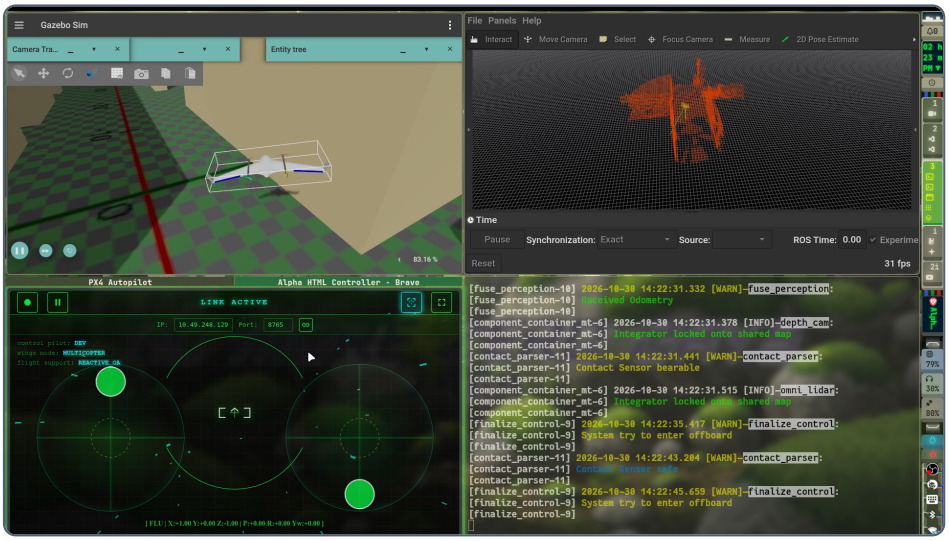
Internship: Cross-Platform ITAM Endpoint Device Agent 16/7 - 17/9/2026
GSOFT SOFTWARE *Ho Chi Minh City*

- Built a lightweight, memory-safe IT-asset monitoring daemon for Linux and Windows in **Rust** (Tokio, Unix domain sockets).
- Split the agent into separate binaries (launcher, worker, fetcher) connected by IPC, so a fetcher crash cannot take down the main daemon; scheduled tasks with a custom min-heap scheduler.
- Implemented OS-specific system calls and the IPC layer; profiled memory and CPU bottlenecks with flamegraph and hyperfine.
- Delivered a working cross-platform MVP with scheduled data fetching and resilient inter-process communication, with no memory leaks observed.

Featured Project

Drone Simulation Pipeline Passion project

Built a ROS2 + PX4 + Gazebo simulation pipeline to train AI models for VTOL acrobatic flight (details on page 2).



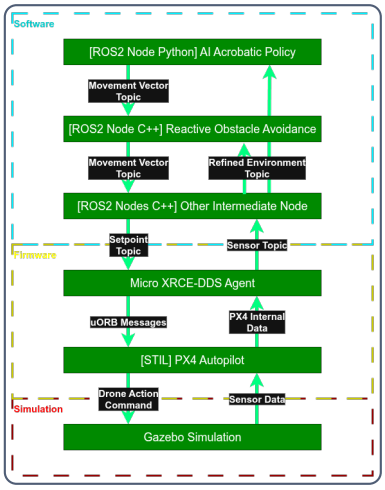
Drone_Simulation_Preview.png

Project 1 Acrobatic Drone Pipeline with Latent Imitation Learning

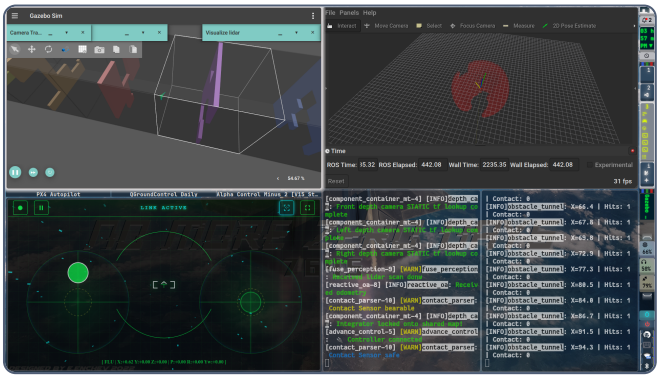
AI-Oriented Multidisciplinary Project 2025

ROS2, PX4, Gazebo Harmonic, Micro XRCE-DDS, Python

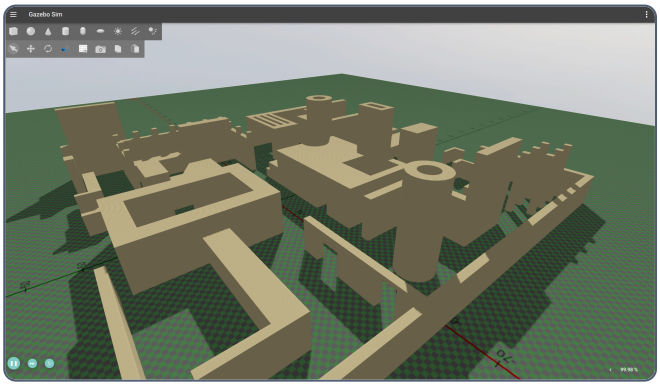
- Built a simulation and control pipeline for an "AI co-pilot" that learns from expert pilot input and turns noisy human commands into safe 3D trajectories.
- Integrated Gazebo Harmonic (physics), PX4 (flight control, 400 Hz), Micro XRCE-DDS (PX4–ROS2 link) and ROS2 (offboard middleware, 80 Hz); the AI runs as a Python ROS2 node at 0.5 Hz behind a safety and obstacle-avoidance layer.
- Collected expert control and telemetry data in a procedurally generated obstacle-tunnel world, trained models offline, and evaluated them offline and in real-time user-controlled flights.
- Validated the non-AI pipeline with autonomous flights of up to 1 km without collisions (average simulated latency 290 µs). Early AI-controlled results were noisy and sometimes conflicted with pilot input, which pointed to design and data-flow issues between the high- and low-level layers.
- Now redesigning the pipeline around low-level data processing, system integration and control structure to support more dynamic maneuvers.



System_Pipeline.png



Test_non-AI_behavior.png



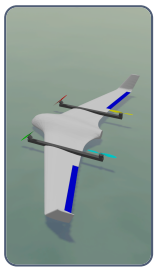
Test_Control_World.png

Project 2 VTOL Acrobatic Flight & Obstacle Avoidance with RL

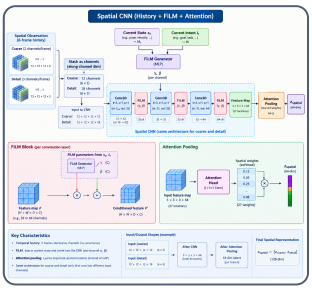
Capstone Project (in progress) 2026

C++, ROS2 Jazzy, PyTorch, Voxblox (ESDF), Gazebo Harmonic, PX4

- Designing a pilot-intent-driven control architecture that gives pilots of any skill level expert-level control of a VTOL while keeping deterministic spatial safety.
- The policy takes pilot or planner intent, PX4-estimated drone state, and a 3D-CNN latent encoding of the Voxblox ESDF map, and outputs normalized actuator commands to the PX4 mixer.
- Staged RL curriculum (hover → tracking → obstacle avoidance), each stage with its own reward and an entropy objective to explore the flight envelope.
- Deterministic reactive fallback using APF path planning for extra safety.



VTOL.png



3D CNN design.png